Naming

① element: the name of element is the same with that of atom

H~Kr: H: Hydrogen He: Helium Li: Lithium Be: Beryllium B: Boron

C: Carbon N: Nitrogen O: Oxygen F: Fluorine Ne: Neon Na: Sodium

Mg: Magnesium Al: Aluminium Si: Silicon P: Phosphorus S: Sulfur Cl: Chlorine

Ar: Argon K: Potassium Ca: Calcium Sc: Scandium Ti: Titanium V: Vanadium

Cr: Chromium Mn: Manganese Fe: Iron Co: Cobalt Ni: Nickel Cu: Copper

Zn: Zinc Ga: Gallium Ge: Germanium As: Arsenic Se: Selenium Br: Bromine

Kr: Krypton

Ag: Silver Sn: Tin I: Iodine Ba: Barium W: Tungsten Pt: Platinum

U: Uranium Au: Gold Hg: Mercury 氕: protium 氘: deuterium

② covalent compound (contain 2 types of atoms) 氚 tritium

Keep the first element the same, the suffix of the second element change into -ide

prefix indicate the number of atom

mono- di- tri- tetra- penta- hexa- hepta- octa- nona- deca-

⚠ "mono-" is usually dropped from the first element

⚠ If the prefix ends in a vowel and the element name begins with a vowel, drop the last vowel in the prefix

⚠ NH_3: ammonia H_2O: water CH_4: methane PH_3: phosphine H_2S: hydrogen sulfide

root of the second element

ox- nitr- fluor- chlor- brom- iod- sulf- phosph- carb- hydr-

③ ionic compound the name of the cation + anion

Cation: (a) NH_4^+ ammonium

(b) the name of the metal cation is the same with metal element

(c) the oxidation number of transition metal needs to be appeared after the name of metal ion

Anion: (a) single-element ions: root + ide

(b) polyatomic ion contain oxygen: root +

per + ~ + ate 高氯酸根

~ + ate 氯酸根

~ + ite 亚氯酸根

hypo + ~ + ite 次氯酸根

ClO_4^- perchlorate

ClO_3^- chlorate

ClO_2^- chlorite

ClO^- hypochlorite

NO_3^-: nitrate PO_4^{3-}: phosphate CO_3^{2-} carbonate

NO_2^-: nitrite

⚠ if a hydrogen is attached to a polyatomic ion, prefix it with __hydrogen

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酸:

由二元化合物得到的无氧酸: hydro-stem-ic acid　　HCl: hydrochloric acid

含氧酸 root+ { per+ ~+ic / ~+ic / ~+ous / hypo+ ~+ous } acid

⚠ H_3PO_4 phosphoric acid
H_3PO_3 phosphorous acid
根变了

碱: cation + hydroxide

多背:　　root变了,同phosphoric acid.

1. H_2SO_4 sulfuric acid
2. HNO_3 nitric acid
3. HCl hydrochloric acid
4. H_3PO_4 phosphoric acid
5. CH_3COOH acetic acid (ethanoic acid)
6. CH_3COO^- acetate (ethanoate)
7. $HClO_4$ perchloric acid
8. $HClO_3$ chloric acid
9. $HClO_2$ chlorous acid
10. HClO hypochlorous acid
11. NaOH sodium hydroxide
12. K_2S potassium sulfide
13. NH_3 ammonia
14. CH_4 methane
15. NH_4Cl ammonium chloride
16. NH_4Br ammonium bromide
17. NH_4I ammonium iodide
18. KH_2PO_4 potassium dihydrogen phosphate
19. $CuSO_4 \cdot 5H_2O$ copper (II) sulfate pentahydrate ☆
20. CO carbon monoxide
21. CO_2 carbon dioxide
22. SO_3 sulfur trioxide
23. CCl_4 carbon tetrachloride
24. CO_3^{2-} carbonate
25. H_2S hydrogen sulfide
26. $KMnO_4$ potassium permanganate
27. BBr_3 boron bromide
28. PbI_2 lead (II) iodide　Pb {+2, +4} oxidation state.
29. Na_2O_2 sodium peroxide
30. $CaCO_3$ calcium carbonate
31. $FeSO_4$ iron(II) sulfate
32. $K_2C_2O_4$ potassium oxalate 草酸钾
33. $Al(OH)_3$ aluminium hydroxide
34. $NaBH_4$ sodium borohydride
35. $LiAlH_4$ lithium aluminium hyaride
36. O_3 ozone
37. $K_2Cr_2O_7$ potassium dichromate

氕: protium
氘: deuterium
氚: tritium

dilute & concentrated

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